

CMPE2004 Advanced Programming — Term Project Report

Project Title

Reminder Assistant (Hatırlatıcı Asistan) — An Accessible Daily Reminder App for Disadvantaged Individuals

Student Information

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Course	CMPE2004 Advanced Programming
Semester	2025–2026 Spring
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Selected Project Option

Option 1: Accessible Daily Reminder App

This option was chosen because daily routines such as taking medication, drinking water, and attending appointments are critical for many disadvantaged individuals. A simple, readable, and notification-based reminder app can provide meaningful support in everyday life.

Purpose of the App

The purpose of Reminder Assistant is to help individuals who need support in remembering daily tasks. The app is designed with accessibility in mind: large buttons, readable text, clear navigation, and notification alerts at scheduled times.

The application aims to be both technically correct and socially meaningful, supporting users who may struggle with memory, routine management, or complex interfaces.

Main Features

1. Daily Reminder Management

- Add reminders with a title and time
- View reminders in a scrollable list on the main screen
- Delete reminders with a long-press confirmation dialog

2. Notification Alerts

- Scheduled reminders trigger system notifications at the selected time
- Uses AlarmManager and a BroadcastReceiver

3. Daily Support Message from the Internet

- Fetches an inspirational quote from <https://api.quotable.io/random>
- Parses JSON and displays the message on the home screen

4. Accessible User Interface

- Large buttons (minimum 70dp height)
- Readable font sizes (18sp–28sp)
- ImageView logo and clear section labels
- Material Design components and custom styles

5. Navigation and Practical Intents

- Explicit intents between three activities
- Options menu for About screen and accessibility information
- Implicit intent to open a WHO accessibility webpage in the browser
- Share reminder text using ACTION_SEND

6. Visual Enhancements

- Custom theme and color palette
- Fade-in animation on main content
- Card-style reminder list items

Technical Structure

Architecture Overview

```
com.example.term_project
├── activity/
│   ├── BaseActivity.java
│   ├── MainActivity.java
│   ├── AddReminderActivity.java
│   └── AboutActivity.java
├── adapter/
│   └── ReminderAdapter.java
├── data/
│   └── ReminderRepository.java
├── model/
│   └── Reminder.java
├── network/
│   ├── NetworkManager.java
│   └── QuoteParser.java
├── receiver/
│   └── ReminderAlarmReceiver.java
└── util/
    ├── ReminderScheduler.java
    └── ReminderNotificationHelper.java
```

Screens (Activities)

Activity	Role
MainActivity	Launcher screen; shows daily quote, reminder list, add button, and menu
AddReminderActivity	Form to enter reminder title and select time via TimePickerDialog
AboutActivity	Displays app purpose, features, and course information

Requirement Mapping

Course Requirement	Implementation
UI Components	TextView, EditText, Button, ImageView, ScrollView, RecyclerView, Menu
Multiple Screens	3 activities
Activity Lifecycle	BaseActivity logs onCreate, onStart, onResume, onPause, onStop, onDestroy
Explicit Intents	Main → AddReminder, Main → About
Implicit Intents	ACTION_VIEW (browser), ACTION_SEND (share)
OOP	Classes, encapsulation, inheritance, interface, exception handling
Internet / HTTP	NetworkManager + HttpURLConnection + JSON quote API
Notification / Broadcast	ReminderAlarmReceiver + AlarmManager + NotificationCompat
Styles / Themes	styles.xml, themes.xml, Material Design 3
Animation	fade_in.xml alpha animation

Key Technologies

- Language: Java 11
- IDE: Android Studio
- Min SDK: 24
- Target SDK: 36
- Libraries: AndroidX AppCompat, Material Components, RecyclerView

Challenges and Solutions

1. Running Network Requests Without Blocking the UI

Challenge: HTTP requests must not run on the main thread.

Solution: NetworkManager uses ExecutorService for background work and Handler to post results back to the UI thread safely.

2. Scheduling Reminders Reliably

Challenge: Reminders must fire at the correct time even when the app is in the background.

Solution: ReminderScheduler uses AlarmManager.setExactAndAllowWhileIdle() with a PendingIntent targeting ReminderAlarmReceiver.

3. Notification Permission on Android 13+

Challenge: Android 13 and above require runtime permission for notifications.

Solution: MainActivity requests POST_NOTIFICATIONS at launch using ActivityResultLauncher.

4. Parsing Internet Data Safely

Challenge: Network responses may be invalid or malformed.

Solution: QuoteParser wraps JSON parsing in a try-catch block and returns a fallback message if parsing fails.

5. Accessible Design for Target Users

Challenge: The app must be easy to use for disadvantaged individuals.

Solution: Large touch targets, high-contrast colors, simple navigation flow, and minimal steps to add a reminder.

Conclusion

Reminder Assistant successfully meets the CMPE2004 term project requirements by combining Java OOP principles, Android components, internet connectivity, and notification support in a socially meaningful application.

The app helps disadvantaged users manage daily tasks through a clean, accessible interface. Features such as scheduled notifications, online motivational messages, and simple screen navigation make the application practical for real-world use.

Future improvements could include persistent storage with SharedPreferences or Room database, multiple reminders per day with repeating alarms, and localization in additional languages.

Presentation Outline (Week 14)

7. Problem Definition — Why daily reminders matter for disadvantaged individuals
8. Target User Group — Elderly users, people with memory difficulties, individuals needing routine support
9. Main Features Demo — Add reminder → list update → notification → daily quote
10. Screen Walkthrough — Main, Add Reminder, About
11. Technical Explanation — Activities, intents, OOP classes, HTTP, BroadcastReceiver
12. Q&A

Replace [Your Full Name] and [Your Student ID] before submission.